

Experimental Study of Transfer Learning on Biomedical Named Entity Recognition (NER) between Clinical Data and Patient Forum for Sustainable Healthcare



Introduction

Accurate diagnosis and treatment depend on reliable information, but misdiagnosis and missing drug side-effect data can hinder care. AI and ML can help by analyzing large medical data, with **Named Entity Recognition (NER)** playing a key role in identifying drugs, symptoms, and side effects. However, NER models often struggle when applied across different types of medical texts.



Clinical datasets use **formal** medical language



Patient forum datasets use more **casual, narrative** expressions



Researchers use **transfer learning** to adapt a model from one dataset to another with only a small amount of new data.

Keywords

Named Entity Recognition (NER), Transfer Learning, Domain Adaptation, Adverse Drug Events (ADE), BioBERT, SDG 3 Good Health and Wellbeing

Goals

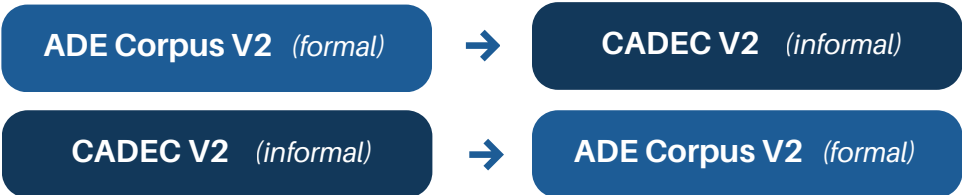
1. Evaluate how **effectively** BERT-based models transfer across domains with different language styles.
2. Determine the **minimum labeled target data** required to reach optimal fine-tuning performance.

Methodology

1. Experimental Scenarios and Models

Cross-Domain Scenarios

Two cross-domain transfer directions were tested:



Models were first **trained on the source domain** and then **fine-tuned on the target domain** using limited data (100, 500, and 1000 samples).

Models BERT BioBERT PubMedBERT

2. Datasets

Dataset	Style	Tokens	Sentences	ADE Entities
ADE Corpus V2	Formal clinical	159,550	6,867	~14k
CADECv2	Informal patient forum	379,566	22,714	~48k

Both datasets use the **BIO** labeling scheme.

3. Training Setup

- Source-domain training: 10 epochs
- Target-domain fine-tuning: 5 epochs
- Batch size: 4
- Learning rate: 2e-5
- Optimizer: AdamW

*No hyperparameter tuning performed.
*Best model selected using **macro F1-score**.

All experiments are conducted using PyTorch + HuggingFace Transformers on Google Colab Pro (NVIDIA T4 GPU)



4. Evaluation

Model performance was measured using:

Entity-level F1-score

Seqeval library

*respects BIO sequence labeling

Performance was compared across:

- ☐ Different **models**
- ☐ **Domain** directions
- ☐ Varying amounts of **target fine-tuning data**

Results & Discussion

Zero-Shot Cross-Domain Performance

Models trained on one domain perform **very poorly** when directly tested on the other without fine-tuning.

ADE → CADEC		CADEC → ADE	
BERT	4.42% F1	BERT	42.15% F1
BioBERT	7.80% F1	BioBERT	48.09% F1
PubMedBERT	1.46% F1	PubMedBERT	40.78% F1

There is a **strong domain shift** between ADE and CADEC. Models fail to generalize across domains due to **differences in language style** (technical vs. conversational), **sentence length and structure**, and **how side effects are expressed** ("insomnia" vs. "I couldn't sleep at all").

Impact of Fine-Tuning With Limited Target Data

Adding even a **small amount of target-domain labeled data** greatly improves model performance.

ADE → CADEC		CADEC → ADE	
Improvement from ~1-7% → ~70% F1		Improvement from ~40-48% → ~63% F1	
BioBERT with 1000 samples	71.35% F1	BioBERT with 1000 samples	63.05% F1
Nearly matches CADEC in-domain score (74.48%)		Close to ADE in-domain score (66.15%)	

Fine-tuning with **100-1000 samples** is enough for models to learn domain-specific expressions and recover most lost performance.



Across all experiments, **BioBERT consistently performs the best** across in-domain, cross-domain, and low-data fine-tuning scenarios.

This is likely because BioBERT was pretrained on large biomedical text sources (PubMed + PMC), giving it stronger medical terminology representations.



Key Findings

- Severe performance drop in zero-shot scenarios highlights the **incompatibility of formal and informal medical language**.
- **Transfer learning is highly effective**, reducing the domain gap with minimal target-labeled data.
- **100-1000 samples** are sufficient to reach near-in-domain performance.
- BioBERT offers the best **generalization and domain adaptation** among the tested models.

Output

#ReMiDi: Web Platform for Medication Recommendation and Diagnosis Based on Machine Learning